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Inventor(s)

Kim; Daejoong et al.

SYSTEM AND A METHOD FOR ESTABLISHING COMMUNICATIONS IN A 5G OPEN RADIO NETWORK (O-RAN)

Abstract

A method for establishing communications between a device and a 5G Open Radio Access Network (O-RAN) includes receiving, at an element management system (EMS) from the device, a connection request including capability information pertaining to the device, transmitting, to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device, receiving, from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server, assigning, based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN, and transmitting, from the EMS to the device, an IP address of the vDU for the device to establish connection with the network.

Inventors: Kim; Daejoong (Seoul, KR), Render; Zeev Vladimir (Ashburn, VA), Chenumolu; Siddhartha (Ashburn, VA)

Applicant: DISH Wireless L.L.C. (Englewood, CO)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a system for establishing communication in an open radio network (O-RAN), and more particularly, a method of establishing communications between a device and a virtual distribution unit (vDU) in the O-RAN.

BACKGROUND

[0002] Network operators for 5G Open Radio Access Networks (O-RANs) often build their radio access network (RAN) by deploying various physical network elements such as radio units (RU), distributed units (DU), and centralized units (CU) in various geographical locations and by connecting them over Internet Protocol (IP) networks. For large and complex RANs, logical associations among network elements are first built in a database, and then the corresponding physical network elements are deployed and instantiated.

SUMMARY

[0003] The present disclosure is directed to establishing communications between a radio unit (RU) and a virtual distribution unit (vDU) in an open radio access network (O-RAN).

[0004] According to one aspect of the subject matter described in this application, a method for establishing communications between a device and a 5G Open Radio Access Network (O-RAN) can include receiving, at an element management system (EMS) from the device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device, transmitting, by the EMS to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device, receiving, at the EMS from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR, assigning, by the EMS based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN, updating, by the EMS, a database to link the device with the one or more identifiers, the IP address of the CSR, and the vDU that can support the device, the database storing information representing connections among network entities of the 5G O-RAN, and transmitting, from the EMS to the device, an IP address of the vDU for the device to establish connection with the network.

[0005] Implementations according to this aspect can include one or more of the following features. For example, the one or more identifiers can include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

[0006] In some implementations, the capability information can identify a carrier component supported by the device, and updating the database can include assigning the cell ID to a cell of the device based on the capability information. In some examples, one or more carrier components can be assigned with one or more cell IDs, and each cell ID can be assigned with a PCID, an IP address of a device, an IP address of a vDU. In some examples, each carrier component can be assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network.

[0007] In some implementations, the device can be a radio unit (RU). In some examples, the connection request includes an IP address of the device.

[0008] According to another aspect of the subject matter described in this application, an element management system (EMS) configured to operate in a 5G Open Radio Access Network (O-RAN) can include at least one processor, and memory coupled to the at least one processor and storing

instructions that, based on being executed by the at least one processor, perform operations. The operations can include receiving, from a device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device, transmitting, to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device, receiving, from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR, assigning, based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN, updating a database to link the device with the one or more identifiers, the IP address of the CSR, and the vDU that can support the device, the database storing information representing connections among network entities of the 5G O-RAN, and transmitting, to the device, an IP address of the vDU for the device to establish connection with the network.

[0009] Implementations according to this aspect can include one or more of the following features. For example, the one or more identifiers can include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

[0010] In some implementations, the capability information can identify a carrier component supported by the device, and updating the database can include assigning the cell ID to a cell of the device based on the capability information. In some examples, one or more carrier components can be assigned with one or more cell IDs, and each cell ID can be assigned with a PCID, an IP address of a device, an IP address of a vDU. In some examples, each carrier component can be assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network.

[0011] In some implementations, the device can be a radio unit (RU). In some examples, the connection request includes an IP address of the device.

[0012] According to another aspect this document describes a non-transitory machine-readable storage device storing a program, wherein execution of the program causes one or more computers of an element management system (EMS) in a 5G Open Radio Access Network (O-RAN) to perform operations including: receiving, from a device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device, transmitting, to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device, receiving, from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR, assigning, based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN to the device, updating a database to link the device with the one or more identifiers and the IP address of the CSR, the database storing information representing connections among network entities of the 5G O-RAN, and transmitting, to the device, an IP address of the vDU for the device to establish connection with the network.

[0013] Implementations according to this aspect can include one or more of the following features. For example, the one or more identifiers can include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

[0014] In some implementations, the capability information can identify a carrier component supported by the device, and updating the database can include assigning the cell ID to a cell of the device based on the capability information. In some examples, one or more carrier components can be assigned with one or more cell IDs, and each cell ID can be assigned with a PCID, an IP address

of a device, an IP address of a vDU. In some implementations, each carrier component can be assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network.

[0015] In some implementations, the device can be a radio unit, and the connection request can include an IP address of the device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a diagram illustrating an example of an open radio access network (O-RAN).

[0017] FIG. 2 is a diagram illustrating an example of operations for establishing communications between a device and a vDU.

[0018] FIG. 3 is a flowchart showing an exemplary process for establishing communications between a device and a vDU.

[0019] FIG. 4 is a flowchart showing an exemplary process for managing failure in communication establishment between a device and a vDU.

[0020] FIG. 5 is a diagram illustrating a computing system that can be used in connection with computer-implemented methods described in this specification.

DETAILED DESCRIPTION

[0021] A radio unit (RU) is a network element that handles various physical layer functionalities such as beamforming. As with other physical network elements in a 5G O-RAN, physical configurations for RUs are often mapped to the database of network elements manually during the RAN database setup. For example, prior to installing an RU at an onsite location, the physical RU is pre-associated with a corresponding logical network in the database—where the serial number or the MAC address of the RU is linked to cell IDs and virtual distribution units (vDUs) of the RAN. Consequently, in the installation process, it is necessary to retrieve the specific RU, potentially from a sizable batch of RUs stored in a warehouse.

[0022] In some cases, this can result in significantly longer delays, as compared to a situation where any readily available RU can be selected for installation. In certain cases, engineers may arrive at a site with an incorrect RU, necessitating either the retrieval of the accurate physical RU or reliance on a potentially time-consuming database update process prior to installation. A similar situation applies for a situation where a correctly-retrieved RU turns out to be defective.

[0023] For these reasons, a single missing or defective RU can cause significant delays in completing site installations, even if a spare RU is readily available. Furthermore, in the event of RU failures, a replacement cannot be seamlessly deployed until the database is updated. In certain scenarios, updating the database may involve additional personnel, resulting in associated logistical delays. This could potentially extend service interruption times and restrict the on-site technician's ability to independently address the issue.

[0024] The present disclosure is directed to address the above-referenced issues by avoiding the above-noted pre-configuration of RUs, but instead allowing RUs to be associated with the RAN database after the physical installation process. RUs can be automatically connected to a vDU that manages the area where the RU is located based on a corresponding cell site router (CSR) through which a message was relayed from the RU to a DHCP server. The IP address, once assigned to a CSR, remains unchanged even if the CSR is rebooted or even if the hardware associated with the CSR is replaced. The present disclosure allows an EMS to associate RUs with the logical network post-installation by communicating through the corresponding CSR. For example, because the location of the CSR is known, the location of the RU can be uniquely identified based on the IP address of the RU being found to be assigned via the CSR. By allowing an RU to be associated

with the logical network based on the IP address of the corresponding CSR rather than the serial number of the RU, the mapping of the RU to the logical network can be postponed until the RU is physically installed at a site. This in turn can allow for any random RU to be physically deployed at a cell site, thereby alleviating the inefficiencies and delays associated with retrieving and installing particular RUs preconfigured for particular physical locations. In some implementations, the concepts described herein can be utilized for other network elements such as DUs and CUs.

[0025] FIG. 1 is a diagram illustrating an example of an open radio access network (O-RAN) **100**. Referring to FIG. 1, the O-RAN **100** can include a plurality of radio unit (RUs) **111-116**, a cell site router (CSR) **110**, a virtual distribution unit (vDU) **120**, a networking infrastructure **130**, and a virtual central unit (vCU) **135**.

[0026] The RU **111** (as well as other RUs **112-116**) can transmit and receive a radio signal in the RAN. The RU **111** can convert digital signals into analog radio signals that are transmitted to and from user devices. In some implementations, the RU **111** can be implemented at cell sites, such as cell towers or small cells, where they connect to antennas to facilitate wireless communication with user devices.

[0027] The RU **111** in the O-RAN can be configured to support open interfaces, allowing interoperability between equipment from different vendors. In some implementations, the RU **111** can support multiple radio access technologies (RATs) and frequency bands, contributing to the scalability and adaptability of the 5G network.

[0028] In some implementations, the RUs **111-113** can be a dual-band RU that can support dual bands. In this context, “dual-band” may refer to a single RU supporting multi cells, where such information is included in capability information of the RU. For example, when an element management system (EMS), which described below in detail, receives the capability information from the RU, the EMS can determine which vDU can support the RU.

[0029] In some implementations, the RUs **114-116** can be a tri-band RU that can support tri-bands. In this context, each band can represent either a single cell or multiple cells, depending on whether the band comprises multiple carrier components.

[0030] The CSR **110** can manage communication between the RU and the vDU. For example, the CSR **110** can relay data transmitted from the plurality of RUs **111-116** to the vDU **120** through the networking infrastructure **130**. The CSR **110** can handle data forwarding, routing, and other networking functions. The CSR **110** can connect the RUs **111-116** at the cell site to the vDU **120**, facilitating the transfer of user data and control signals. The CSR can support standard networking protocols, such as IP (Internet Protocol), and ensure the efficient and secure flow of data between the cell site and the vDU.

[0031] The IP address, once assigned to the CSR **110**, remains unchanged even if the CSR is rebooted or even if the hardware associated with the CSR **110** is replaced. Thus, the IP address of the CSR **110** can be used as an indicator of a specific location in the O-RAN **100**. In some implementations, the CSR can be located in every site where the RUs are installed.

[0032] A cell site can be divided into three sectors, Alpha, Beta, and Gamma. These sectors can refer to different directional coverage areas within the cell site, each covering a portion of the total area. In some implementations, these sectors can be randomly selected such that the assignment of these names to the sectors is not based on a predetermined or systematic but can be done in a random manner.

[0033] A cell ID (Cell Identity) and a PCID (Physical Cell ID) can refer to identifiers associated with each cell. The cell ID can uniquely identify a cell within the network, while the PCID can be used to differentiate between cells in the area. In some implementations, the arrangement of the cell ID and the PCID can be done randomly among the three sectors belonging to the same site, where this random arrangement can ensure that neighboring sectors have distinct identifiers.

[0034] The use of random selection for sector names (alpha, beta, gamma) and the random arrangement of cell ID and PCID among sectors within the same site can contribute to a flexible

and varied naming and identification scheme. This randomness can be advantageous in scenarios where predictability or specific patterns are not necessary, providing a degree of flexibility in network planning and management.

[0035] The vDU **120** may refer to a virtualized component of the distributed unit in the 5G network architecture. The vDU **120** can process user plane data, including functions related to radio resource management and data forwarding. In some implementations, the vDU **120** can be interconnected with the radio units **111-116** and the vCU **135**, forming a distributed and virtualized network architecture that enhances resource utilization and network efficiency.

[0036] In some implementations, the connection between the CSR **110** and the vDU **120** can be established over the network infrastructure **130**. For example, the connection can be based on protocols such as Ethernet.

[0037] The vCU **135** may refer to a virtual central unit that can process and manage radio resources. Some of key operations performed by the vCU **135** can include, radio resource management, signal processing, coordination with the vDU **120**, and supporting network slicing.

[0038] FIG. **2** is a diagram illustrating an example of operations for establishing communications between a device (e.g., the RU **111**) and a vDU **120**.

[0039] In step **S201**, the RU can be installed at a site and powered on. At this initial state, the RU does not have a preconfigured IP address such that the RU needs to obtain an IP address.

[0040] In step **S202**, the RU can initiate a dynamic host configuration protocol (DHCP) process by sending a DHCP Discovery message. This message is broadcasted to the network **100**, seeking a DHCP server **140**.

[0041] The CSR **110**, acting as a router and DHCP relay agent, can intercept the broadcasted DHCP Discovery message from the RU and, in step **S203**, can forward the DHCP Discovery message to the DHCP server **140**.

[0042] The DHCP server **140** can receive the DHCP Discovery message and process the request by identifying available IP addresses.

[0043] In step **S204**, the DHCP server **140** can respond to the CSR **110** with a DHCP Offer message, which includes details such as the suggested IP address for the RU and other configuration parameters.

[0044] In step **S205**, the CSR **110** can forward the DHCP Offer to the RU **111**, and the RU **111** can receive the offer and considers the suggested IP address.

[0045] In some implementations, the RU **111**, having received one or more DHCP Offer messages, can select an offered IP address. In step **S206**, the RU **111** can transmit a DHCP Request message, broadcasting its request for the selected IP address.

[0046] In step **S207**, the CSR **110** can intercept the DHCP request message and forward the DHCP Request message to the DHCP server **140**, indicating that the RU has accepted the offered IP address.

[0047] In step **S208**, the DHCP server **140** can receive the DHCP Request and acknowledge it by sending a DHCP Acknowledgment (ACK) message to the CSR **110**.

[0048] In step **S209**, the CSR **110** can receive the DHCP ACK and forward the DHCP ACK to the RU **111**.

[0049] The RU **111** can receive the DHCP Acknowledgment, and, in step **S210**, can transmit a connect request to an element management system (EMS) **150** to establish communication with the network. In some implementations, the connection request can include capability information pertaining to the device, which identifies a carrier component supported by the device. In some implementations, the connection request can include at least one of an IP address of the RU **111** or capability information regarding the bands, carrier components, etc.

[0050] The EMS **150** can manage and monitor individual network elements within the network **100**. The EMS **150** can provide a centralized platform for configuring, monitoring, and maintaining these individual elements.

[0051] In step **S211**, the EMS **150** can transmit, to the DHCP server **140** associated with the RU **111**, a request for an Internet Protocol (IP) address associated with the RU **111**. In some implementations, the request can include RU's IP address.

[0052] In step **212**, the DHCP server can, in response to the request received in step **S211**, transmit an IP address of the CSR **110**, which is deployed between the RU **111** and the DHCP server **140** such that communications between the device and the DHCP server are routed through the CSR. The DHCP server can identify the IP address of the CSR **110** by referring to a table that specifies which IP addresses are assigned to specific CSRs.

[0053] In step **S213**, the EMS **150** can assign, based at least in part on the capability information pertaining to the RU **111**, (i) one or more identifiers and (ii) a vDU **120** associated with the 5G O-RAN **100**.

[0054] In some implementations, the logical network encompassing the virtual representation of vDUs, cell IDs, and PCIDs within the 5G network can be built in a database **160** when the RU **111** is implemented at the site. For example, the logical network can include information regarding a site, a number of sectors at the site, one or more CSRs implemented in the site, a location of each of the one or more CSRs, and one or more cell IDs, PCIDs, and vDUs associated to the site. In some implementations, a number of cell IDs can correspond to a number of sectors.

[0055] Using this logical network, the EMS can associate the RU **111** to the cell IDs, the PCIDs, and a vDU.

[0056] For example, the one or more identifiers include a cell identifier (cell ID) and a physical cell identifier (PCID), where the cell ID is a specific identifier assigned to a cell within in a remote site, and the PCID is a specific identifier indicating a physical layer identity of the cell, as described above with respect to FIG. **1**. In some implementations, the cell ID can be assigned to a cell of the RU **111**.

[0057] The EMS **150** can also associate the vDU **120** that is connected to the CSR **110**. As the vDU is created, information regarding a list of remote sites that the vDU should cover and a list of RUs in each site is available. Through the operations discussed above, when the EMS receives a connection request from the RU **111** in step **S210**, the EMS **150** asks the DHCP server **140** which CSR is involved to assign the IP address to the RU. The knowledge on the CSR can enable that EMS to select a vDU that will serve the RU **111**.

[0058] In step **S214**, the EMS **150** can save the configuration related to the assignment made in step **S213** in the database **160** to update the logical network. For example, based on the capability information identifying a specific carrier component supported by the RU **111**, the logical network in the database is updated by assigning the cell ID associated with the specific carrier component to a cell of the RU **111**.

[0059] In some implementations, one or more carrier components in the site can be assigned with one or more cell IDs where each cell ID can be assigned with a PCID, an IP address of a RU, an IP address of a vDU. For example, the EMS **150** can update the logical network by assigning the IP address of the RU **111** for the cell ID that is associated with the carrier component supported by the RU, as indicated in the capability information.

[0060] In some implementations, each carrier component is assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network **100**.

[0061] In step **S215**, the EMS **150** can transmit, to the RU **111**, an IP address of the vDU **120** for the RU **111** to establish connection with the vDU **120**.

[0062] FIG. **3** is flowchart showing an exemplary process **300** for establishing communications between a device and a vDU.

[0063] In step **310**, an EMS **150** can receive a connect request from a RU **111** to establish communication with the network **100**. In some implementations, the connection request can include capability information pertaining to the device, which identifies a carrier component

supported by the device.

[0064] In step **S320**, the EMS **150** can transmit, to the DHCP server **140** associated with the RU **111**, a request for an Internet Protocol (IP) address of a CSR associated with the RU **111**.

[0065] In step **330**, the EMS **150** can receive, in response to the request sent in step **S320**, an IP address of the CSR **110**, which is deployed between the RU **111** and the DHCP server **140** such that communications between the device and the DHCP server are routed through the CSR. s

[0066] In step **S340**, the EMS **150** can assign, based at least in part on the capability information pertaining to the RU **111**, (i) one or more identifiers and (ii) a vDU **120** associated with the 5G O-RAN **100**.

[0067] In some implementations, the logical network encompassing the virtual representation of vDUs, cell IDs, and PCIDs within the 5G network can be built in a database **160** when the RU **111** is implemented at the site. For example, the logical network can include information regarding a site, a number of sectors at the site, one or more CSRs implemented in the site, a location of each of the one or more CSRs, and one or more cell IDs, PCIDs, and vDUs associated to the site.

[0068] Using this logical network, the EMS can associate the RU **111** to the cell IDs, the PCIDs, and a vDU.

[0069] For example, the one or more identifiers include a cell identifier (cell ID) and a physical cell identifier (PCID), where the cell ID is a specific identifier assigned to a cell within in a remote site associated with the CSR, and the PCID is a specific identifier indicating a physical layer identity of the cell, as described above with respect to FIG. **1**. In some implementations, the cell ID can be assigned to a cell of the RU **111**.

[0070] The EMS **150** can also associate the vDU **120** that is connected to the CSR **110**. As the vDU is created, information regarding a list of remote sites that the vDU should cover and a list of RUs in each site is available. Through the operations discussed above, when the EMS receives a connection request from the RU **111** in step **S210**, the EMS **150** asks the DHCP server **140** which CSR is involved to assign the IP address to the RU. The knowledge on the CSR can enable that EMS to select a vDU that will serve the RU **111**.

[0071] In step **S350**, the EMS **150** can save the configuration related to the assignment made in step **S340** in the database **160** to update the logical network. For example, based on the capability information identifying a specific carrier component supported by the RU **111**, the logical network in the database is updated by assigning the cell ID associated with the specific carrier component to a cell of the RU **111**.

[0072] In some implementations, one or more carrier components in the site can be assigned with one or more cell IDs where each cell ID can be assigned with a PCID, an IP address of a RU, an IP address of a vDU. For example, the EMS **150** can update the logical network by assigning the IP address of the RU **111** for the cell ID that is associated with the carrier component supported by the RU, as indicated in the capability information.

[0073] In some implementations, each carrier component is assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network **100**.

[0074] In step **S360**, the EMS **150** can transmit, to the RU **111**, an IP address of the vDU **120** for the RU **111** to establish connection with the vDU **120**.

[0075] FIG. **4** is flowchart showing an exemplary process **500** for managing failure in communication establishment between a device and a vDU.

[0076] In step **510**, an EMS **150** can determine that a unique combination of the one or more identifiers is not available for assignment to the RU **111**. For example, if all cell IDs are assigned, the EMS **150** can determine that the RU **111** may not be connected to the vDU **120**.

[0077] In step **520**, the EMS **150** can transmit a connection failure message to the RU **111** based on a determination in step **510** that the unique combination of the one or more identifiers is not available for assignment to the RU **111**.

[0078] FIG. 5 shows an example of a computing device **400** and a mobile computing device **450** (also referred to herein as a wireless device) that are employed to execute implementations of the present disclosure. The computing device **400** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The mobile computing device **450** is intended to represent various forms of mobile devices, such as personal digital assistants, cellular telephones, smart-phones, AR devices, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be examples only, and are not meant to be limiting. The computing device **400** and/or the mobile computing device **450** can form at least a portion of the application installation environment described above.

[0079] The computing device **400** includes a processor **402**, a memory **404**, a storage device **406**, a high-speed interface **408**, and a low-speed interface **412**. In some implementations, the high-speed interface **408** connects to the memory **404** and multiple high-speed expansion ports **410**. In some implementations, the low-speed interface **412** connects to a low-speed expansion port **414** and the storage device **406**. Each of the processor **402**, the memory **404**, the storage device **406**, the high-speed interface **408**, the high-speed expansion ports **410**, and the low-speed interface **412**, are interconnected using various buses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **402** can process instructions for execution within the computing device **400**, including instructions stored in the memory **404** and/or on the storage device **406** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as a display **416** coupled to the high-speed interface **408**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. In addition, multiple computing devices may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

[0080] The memory **404** stores information within the computing device **400**. In some implementations, the memory **404** is a volatile memory unit or units. In some implementations, the memory **404** is a non-volatile memory unit or units. The memory **404** may also be another form of a computer-readable medium, such as a magnetic or optical disk.

[0081] The storage device **406** is capable of providing mass storage for the computing device **400**. In some implementations, the storage device **406** may be or include a computer-readable medium, such as a floppy disk device, a hard disk device, an optical disk device, a tape device, a flash memory, or other similar solid-state memory device, or an array of devices, including devices in a storage area network or other configurations. Instructions can be stored in an information carrier. The instructions, when executed by one or more processing devices, such as processor **402**, perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices, such as computer-readable or machine-readable mediums, such as the memory **404**, the storage device **406**, or memory on the processor **402**.

[0082] The high-speed interface **408** manages bandwidth-intensive operations for the computing device **400**, while the low-speed interface **412** manages lower bandwidth-intensive operations. Such allocation of functions is an example only. In some implementations, the high-speed interface **408** is coupled to the memory **404**, the display **416** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **410**, which may accept various expansion cards. In the implementation, the low-speed interface **412** is coupled to the storage device **406** and the low-speed expansion port **414**. The low-speed expansion port **414**, which may include various communication ports (e.g., Universal Serial Bus (USB), Bluetooth, Ethernet, wireless Ethernet) may be coupled to one or more input/output devices. Such input/output devices may include a scanner, a printing device, or a keyboard or mouse. The input/output devices may also be coupled to the low-speed expansion port **414** through a network adapter. Such network input/output devices may include, for example, a switch or router.

[0083] The computing device **400** may be implemented in a number of different forms, as shown in the FIG. 5. For example, it may be implemented as a standard server **420**, or multiple times in a group of such servers. In addition, it may be implemented in a personal computer such as a laptop computer **422**. It may also be implemented as part of a rack server system **424**. Alternatively, components from the computing device **400** may be combined with other components in a mobile device, such as a mobile computing device **450**. Each of such devices may contain one or more of the computing device **400** and the mobile computing device **450**, and an entire system may be made up of multiple computing devices communicating with each other. The computing device **400** may be implemented in the RUs **111-116**, the CSR **110**, the vDU **120**, the network infrastructure **130**, the DHCP server **140**, the EMS **150**, and the database **160**, described with respect to FIGS. 1-4.

[0084] The mobile computing device **450** includes a processor **452**; a memory **464**; an input/output device, such as a display **454**; a communication interface **466**; and a transceiver **468**; among other components. The mobile computing device **450** may also be provided with a storage device, such as a micro-drive or other device, to provide additional storage. Each of the processor **452**, the memory **464**, the display **454**, the communication interface **466**, and the transceiver **468**, are interconnected using various buses, and several of the components may be mounted on a common motherboard or in other manners as appropriate. In some implementations, the mobile computing device **450** may include a camera device(s) (not shown).

[0085] The processor **452** can execute instructions within the mobile computing device **450**, including instructions stored in the memory **464**. The processor **452** may be implemented as a chipset of chips that include separate and multiple analog and digital processors. For example, the processor **452** may be a Complex Instruction Set Computers (CISC) processor, a Reduced Instruction Set Computer (RISC) processor, or a Minimal Instruction Set Computer (MISC) processor. The processor **452** may provide, for example, for coordination of the other components of the mobile computing device **450**, such as control of user interfaces (UIs), applications run by the mobile computing device **450**, and/or wireless communication by the mobile computing device **450**.

[0086] The processor **452** may communicate with a user through a control interface **458** and a display interface **456** coupled to the display **454**. The display **454** may be, for example, a Thin-Film-Transistor Liquid Crystal Display (TFT) display, an Organic Light Emitting Diode (OLED) display, or other appropriate display technology. The display interface **456** may include appropriate circuitry for driving the display **454** to present graphical and other information to a user. The control interface **458** may receive commands from a user and convert them for submission to the processor **452**. In addition, an external interface **462** may provide communication with the processor **452**, so as to enable near area communication of the mobile computing device **450** with other devices. The external interface **462** may provide, for example, for wired communication in some implementations, or for wireless communication in other implementations, and multiple interfaces may also be used.

[0087] The memory **464** stores information within the mobile computing device **450**. The memory **464** can be implemented as one or more of a computer-readable medium or media, a volatile memory unit or units, or a non-volatile memory unit or units. An expansion memory **474** may also be provided and connected to the mobile computing device **450** through an expansion interface **472**, which may include, for example, a Single in Line Memory Module (SIMM) card interface. The expansion memory **474** may provide extra storage space for the mobile computing device **450**, or may also store applications or other information for the mobile computing device **450**. Specifically, the expansion memory **474** may include instructions to carry out or supplement the processes described above, and may include secure information also. Thus, for example, the expansion memory **474** may be provided as a security module for the mobile computing device **450**, and may be programmed with instructions that permit secure use of the mobile computing

device **450**. In addition, secure applications may be provided via the SIMM cards, along with additional information, such as placing identifying information on the SIMM card in a non-hackable manner.

[0088] The memory may include, for example, flash memory and/or non-volatile random access memory (NVRAM), as discussed below. In some implementations, instructions are stored in an information carrier. The instructions, when executed by one or more processing devices, such as processor **452**, perform one or more methods, such as those described above. The instructions can also be stored by one or more storage devices, such as one or more computer-readable or machine-readable mediums, such as the memory **464**, the expansion memory **474**, or memory on the processor **452**. In some implementations, the instructions can be received in a propagated signal, such as, over the transceiver **468** or the external interface **462**.

[0089] The mobile computing device **450** may communicate wirelessly through the communication interface **466**, which may include digital signal processing circuitry where necessary. The communication interface **466** may provide for communications under various modes or protocols, such as Global System for Mobile communications (GSM) voice calls, Short Message Service (SMS), Enhanced Messaging Service (EMS), Multimedia Messaging Service (MMS) messaging, code division multiple access (CDMA), time division multiple access (TDMA), Personal Digital Cellular (PDC), Wideband Code Division Multiple Access (WCDMA), CDMA2000, General Packet Radio Service (GPRS). Such communication may occur, for example, through the transceiver **468** using a radio frequency. In addition, short-range communication, such as using a Bluetooth or Wi-Fi, may occur. In addition, a Global Positioning System (GPS) receiver module **470** may provide additional navigation- and location-related wireless data to the mobile computing device **450**, which may be used as appropriate by applications running on the mobile computing device **450**.

[0090] The mobile computing device **450** may also communicate audibly using an audio codec **460**, which may receive spoken information from a user and convert it to usable digital information. The audio codec **460** may likewise generate audible sound for a user, such as through a speaker, e.g., in a handset of the mobile computing device **450**. Such sound may include sound from voice telephone calls, may include recorded sound (e.g., voice messages, music files, etc.) and may also include sound generated by applications operating on the mobile computing device **450**.

[0091] The mobile computing device **450** may be implemented in a number of different forms, as shown in FIG. 5. For example, it may be implemented in the RUs **111-116**, a DU, and a CU described with respect to FIGS. 1-4. Other implementations may include a phone device **482** and a tablet device **484**. The mobile computing device **450** may also be implemented as a component of a smart-phone, personal digital assistant, AR device, or other similar mobile device.

[0092] Computing device **400** and/or **450** can also include USB flash drives. The USB flash drives may store operating systems and other applications. The USB flash drives can include input/output components, such as a wireless transmitter or USB connector that may be inserted into a USB port of another computing device.

[0093] Although a few implementations have been described in detail above, other modifications may be made without departing from the scope of the inventive concepts described herein, and, accordingly, other implementations are within the scope of the following claims.

Claims

1. A method for establishing communications between a device and a 5G Open Radio Access Network (O-RAN), the method comprising: receiving, at an element management system (EMS) from the device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device; transmitting, by the EMS to a dynamic host configuration protocol (DHCP) server associated with the device, a

request for an Internet Protocol (IP) address associated with the device; receiving, at the EMS from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR; assigning, to the device by the EMS based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN; updating, by the EMS, a database to link the device with the one or more identifiers and the IP address of the CSR, the database storing information representing connections among network entities of the 5G O-RAN; and transmitting, from the EMS to the device, an IP address of the vDU for the device to establish connection with the network.

2. The method of claim 1, wherein the one or more identifiers include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

3. The method of claim 2, wherein the capability information identifies a carrier component supported by the device, and wherein updating the database comprises assigning the cell ID to a cell of the device based on and the capability information.

4. The method of claim 3, wherein one or more carrier components are assigned with one or more cell IDs, and wherein each cell ID is assigned with a PCID, an IP address of a device, an IP address of a vDU.

5. The method of claim 4, wherein each carrier component is assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network.

6. The method of claim 1, wherein the device is a radio unit.

7. The method of claim 1, wherein the connection request includes an IP address of the device.

8. An element management system (EMS) configured to operate in a 5G Open Radio Access Network (O-RAN), the system comprising: at least one processor; and memory coupled to the at least one processor and storing instructions that, based on being executed by the at least one processor, perform operations comprising: receiving, from a device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device; transmitting, to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device; receiving, from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR; assigning, based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN to the device; updating a database to link the device with the one or more identifiers and the IP address of the CSR, the database storing information representing connections among network entities of the 5G O-RAN; transmitting, to the device, an IP address of the vDU for the device to establish connection with the network.

9. The system of claim 8, wherein the one or more identifiers include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

10. The system of claim 9, wherein the capability information identifies a carrier component supported by the device, and wherein updating the database comprises assigning the cell ID to a cell of the device based on the capability information.

11. The system of claim 10, wherein one or more carrier components are assigned with one or more cell IDs, and wherein each cell ID is assigned with a PCID, an IP address of a device, an IP address of a vDU.

12. The system of claim 11, wherein each carrier component is assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel

number (ARFCN) identifying a radio frequency carrier in the network.

13. The system of claim 8, wherein the device is a radio unit.

14. The system of claim 8, wherein the connection request includes an IP address of the device.

15. A non-transitory machine-readable storage device storing a program, which upon execution causes one or more computers of an element management system (EMS) in a 5G Open Radio Access Network (O-RAN) to perform operations comprising: receiving, from a device, a connection request for the device to establish communication with the network, the connection request including capability information pertaining to the device; transmitting, to a dynamic host configuration protocol (DHCP) server associated with the device, a request for an Internet Protocol (IP) address associated with the device; receiving, from the DHCP server in response to the request, an IP address of a cell site router (CSR) deployed between the device and the DHCP server such that communications between the device and the DHCP server are routed through the CSR; assigning, based at least in part on the capability information pertaining to the device, (i) one or more identifiers and (ii) a virtual distribution unit (vDU) associated with the 5G O-RAN to the device; updating a database to link the device with the one or more identifiers and the IP address of the CSR, the database storing information representing connections among network entities of the 5G O-RAN; and transmitting, to the device, an IP address of the vDU for the device to establish connection with the network.

16. The non-transitory machine-readable storage device of claim 15, wherein the one or more identifiers include a cell identifier (cell ID) and a physical cell identifier (PCID), the cell ID being a specific identifier assigned to a cell in a remote site, and the PCID being a specific identifier indicating a physical layer identity of the cell.

17. The non-transitory machine-readable storage device of claim 16, wherein the capability information identifies a carrier component supported by the device, and wherein updating the database comprises assigning the cell ID to a cell of the device based on the capability information.

18. The non-transitory machine-readable storage device of claim 17, wherein one or more carrier components are assigned with one or more cell IDs, and wherein each cell ID is assigned with a PCID, an IP address of a device, an IP address of a vDU.

19. The non-transitory machine-readable storage device of claim 18, wherein each carrier component is assigned with at least one uplink (UL) frequency, at least one downlink (DL) frequency, and an absolute radio frequency channel number (ARFCN) identifying a radio frequency carrier in the network.

20. The non-transitory machine-readable storage device of claim 15, wherein the device is a radio unit, and wherein the connection request includes an IP address of the device.
